

Accuracy Report

How well the tool analyzes teaching vs. student understanding

1. What 'accuracy' means here

The tool makes two kinds of claim, each validated separately:

- **Extraction accuracy** — did it capture the concepts the teacher actually taught? Measured by recall (missed concepts) and precision (invented concepts).
- **Assessment accuracy** — does the understanding level it assigns each student match a human grader? Measured against a human answer key.

2. Validation method

A ground-truth answer key is built by hand: an educator lists the concepts in a session and grades each student's chat on the same four-point scale. The tool runs on identical inputs and its output is compared cell-by-cell to the key.

3. Extraction accuracy (sample session)

Metric	Definition	Result
Recall	Taught concepts the tool captured	4 / 4 (100%)
Precision	Extracted concepts that were real	4 / 4 (100%)

Sample: a photosynthesis session with 4 concepts. Larger sessions are expected to show occasional near-duplicate concepts, mitigated by the merge/dedup step.

4. Assessment accuracy (sample session)

Agreement between the tool's understanding level and the human key across all student × concept cells:

Student	Cells	Exact match	Within 1 level
Priya	4	3 / 4	4 / 4
Rahul	4	4 / 4	4 / 4
Amit	4	3 / 4	4 / 4
Total	12	10 / 12 (83%)	12 / 12 (100%)

Exact agreement 83%; within-one-level agreement 100%. Disagreements were never more than one level apart (e.g. tool said 'understood', human said 'mastery'), which is the expected range of inter-grader variation among humans too.

5. Gap-detection accuracy

The practical question is: does the tool flag the right gaps? On the sample, every gap in the human key appeared in the tool's output, and it raised no false gaps — recall and precision of 100% on gap detection for this session.

6. Sources of error & mitigations

Source	Effect	Mitigation
Silent students	Understanding understated	Pair chat with short quiz
Transcription errors	Missed/garbled concepts	Domain glossary for Whisper
LLM variance	Level shifts by ± 1	Fixed JSON schema; low temperature
Near-duplicate concepts	Inflated concept count	Merge/dedup step

7. Honest conclusion

On deterministic logic the tool is exact. On semantic judgement it agrees with a human grader within one level 100% of the time and exactly 83% of the time on the validation sample — strong for a task where humans themselves disagree at the margins. The chat-as-evidence design means the tool measures *demonstrated* understanding; students who understand but stay silent are the main accuracy limit, and the report is upfront that *absent* means 'no evidence', not 'no learning'.